Qassim University
College of Computer | كلية الحاسب
جامعة القصيم

عسّاس

VISION 2030 رؤية
المملكة العربية السعودية
KINGDOM OF SAUDI ARABIA

AI-Driven Road Damage Detection, Prioritization, and Prediction System

Introduction

Road infrastructure deterioration poses significant risks to public safety and increases maintenance costs due to delays and limitations in traditional inspection methods. ASSAS is an AI-powered system designed to improve road maintenance through automated road damage detection, maintenance prioritization, and predictive analysis using the YOLOv8m deep learning model. The system analyzes real-time road imagery captured through municipal vehicle-mounted cameras to detect potholes, cracks, water accumulation, and normal road conditions using computer vision and deep learning techniques, with results displayed through an interactive dashboard that supports maintenance monitoring and decision-making. By integrating detection, prioritization, and prediction into a unified framework, ASSAS enhances operational efficiency and supports smarter and more sustainable road management aligned with Saudi Vision 2030.

Problem and Objective

Traditional road inspection methods are inefficient, time-consuming, and reactive, resulting in delayed damage detection, increased maintenance costs, and compromised road safety.

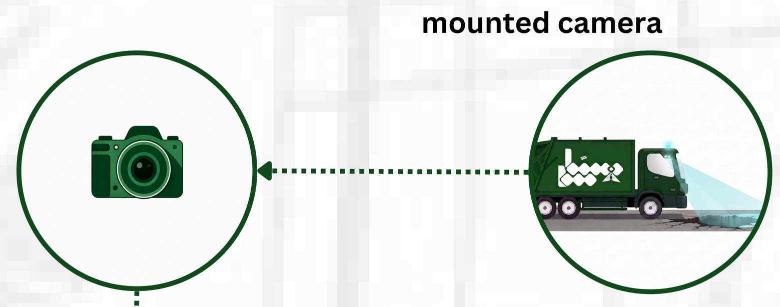
Objectives:

- Develop an AI-driven system for automated road damage detection.
- Assess damage severity to determine maintenance priority.
- Predict future road condition based on detected damage.
- Provide an interactive dashboard for real-time monitoring.
- Support faster decision-making and maintenance planning.

Methodology

1 Image Capture

- Capture road images using vehicle-mounted camera



2 AI Utilization (YOLOv8)

Detect damage type

Pothole | Crack | Water accumulation | Normal

Determine priority (Severity level)

High | Medium | Low | Normal

Predict condition

Critical | Will Deteriorate | May Deteriorate | Stable

Locate damage on map

High Risk | Medium Risk | Low Risk | Normal

3 Generate Report

- Generate case automatically
- Store results in system
- Display in dashboard (Map + Table)

4 Maintenance Action

- Assign case to engineer
- Track status:
  - Not Completed
  - In Progress
  - Completed
- Update after repair completion

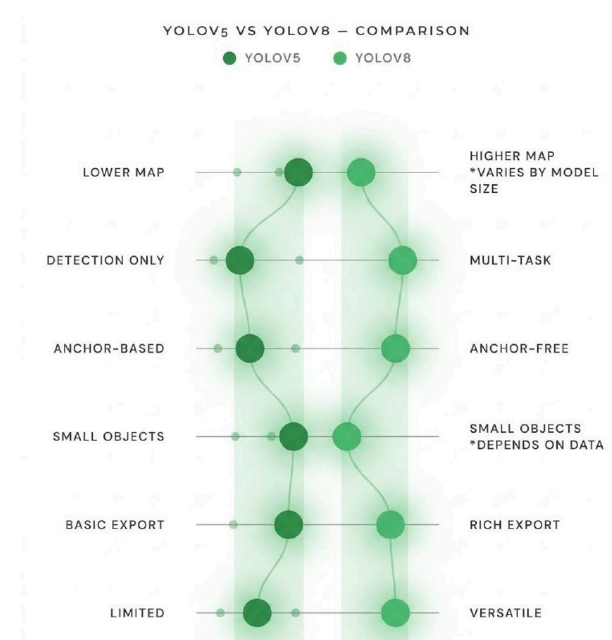
Determine priority (Severity level)

Severity is determined based on the detected **damage type** and its relative size within the image. **Area Ratio** analysis is used to measure detected damage relative to the road image size, while supporting **multi-damage analysis** and identifying the **highest-priority case** within the image.

Predict condition

Predictive analysis estimates the potential **future condition** of detected road damage. Prediction results are displayed through the **dashboard** to support maintenance monitoring and **decision-making**, while supporting **multi-damage prediction analysis** by identifying the highest-priority **prediction state** within the image.

Experiments and Results



Data Preparation

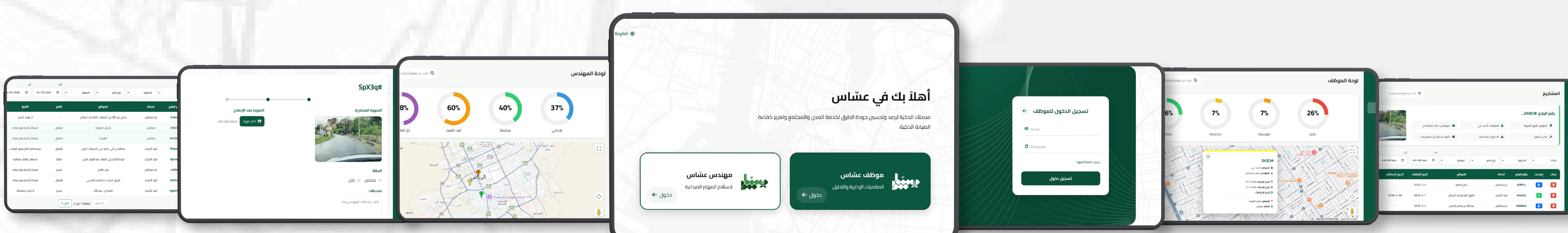
1. **COLLECTION**
   Collected road damage images from multiple sources:
   - Vehicle perspective (municipal cars)
   - Roboflow dataset
   - Google Images (Kaggle)
2. **CLEANING**
   Removed blurry, irrelevant images and duplicates to improve data quality.
3. **ANNOTATION**
   Annotated images using Roboflow for accurate bounding boxes.
4. **AUGMENTATION**
   Applied augmentation (rotation, flipping, brightness, contrast, etc.) to improve model generalization.

Per-Class Detection Performance

Class	Instances	Precision	Recall	mAP@50
CRACKS	527 instances	90.9%	85.0%	91.2%
NORMAL	303 instances	95.0%	97.4%	97.7%
POTHOLES	571 instances	90.2%	83.9%	90.9%
WATER	568 instances	81.0%	74.5%	82.6%

YOLOv8 90.6% OVERALL MAP@50

Total Images : 23,000 +



Tools and Technologies

Figma | HTML5 | roboflow | Postman | CSS | GitHub | Visual Studio Code

Conclusion

ASSAS provides a practical AI-driven approach for improving road maintenance efficiency through faster damage monitoring and maintenance support. Using YOLOv8m, the system achieved 90.6% mAP detection performance on a dataset exceeding 23,000 road images while supporting maintenance prioritization, predictive analysis, monitoring, and interactive case management through a unified dashboard. Future development includes integrating road cameras for real-time monitoring, incorporating weather and traffic data for enhanced prediction, developing more advanced deterioration prediction models, and implementing an integrated field application for employees and engineers. By reducing dependence on manual inspection methods, ASSAS supports data-driven maintenance operations and contributes to developing smarter infrastructure and future smart city solutions aligned with Saudi Vision 2030.

Project Team & Supervisor

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